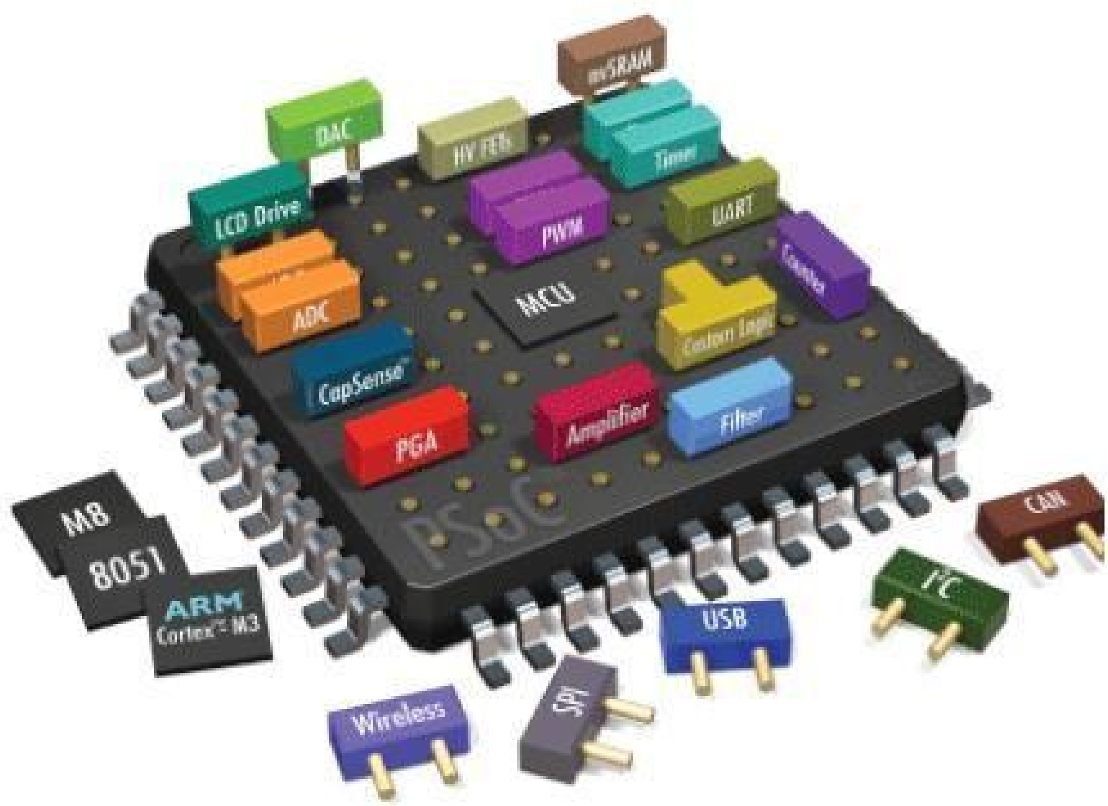
HITEC UNIVERSITY, TAXILA

DEPARTMENT OF COMPUTER ENGINEERING

Lab # 05: Led Pattern generation using EdSim51 Simulator

Embedded Systems

Led Pattern generation using EdSim51 Simulator

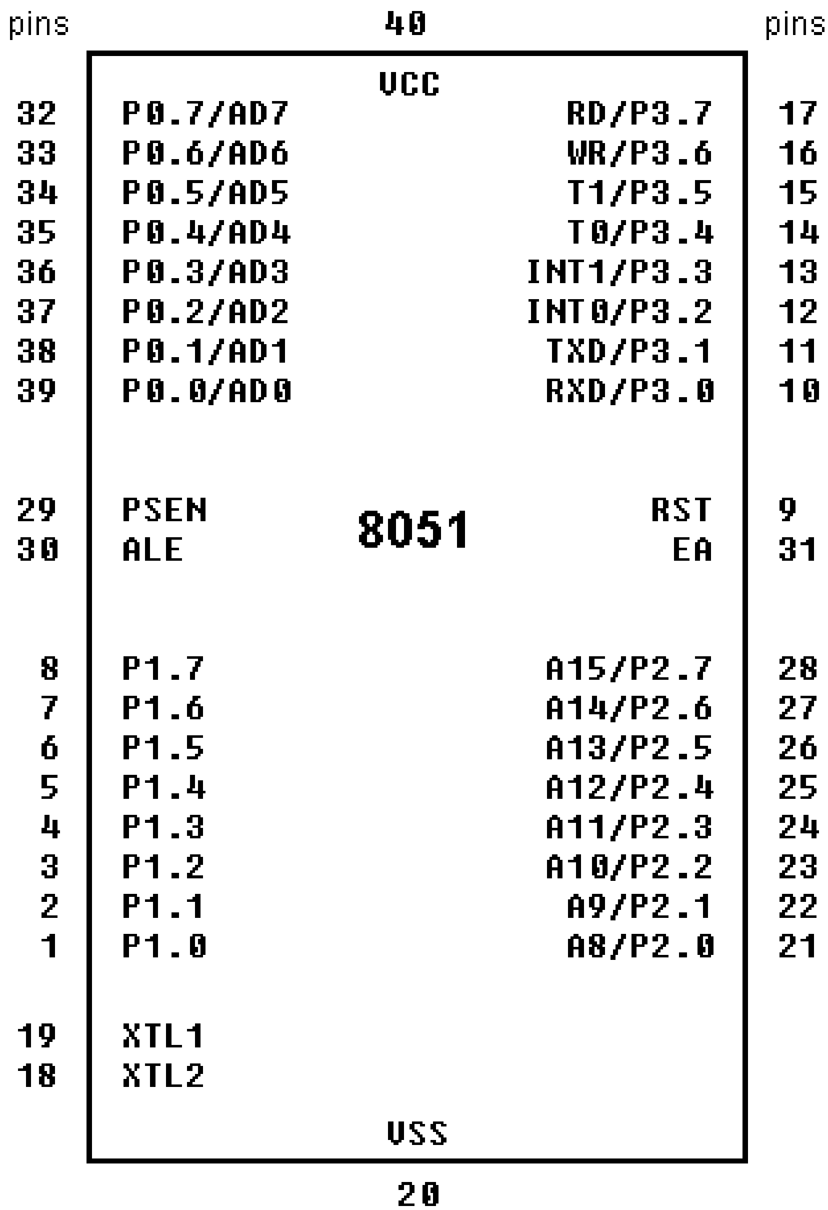


Dated:

Week 05

8051 microcontroller:

8051 microcontroller is designed by Intel in 1981. It is an 8-bit microcontroller. It is built with 40 pins DIP (dual inline package), 4kb of ROM storage and 128 bytes of RAM storage, 2 16-bit timers. It consists of are four parallel 8-bit ports, which are programmable as well as addressable as per the requirement. An on-chip crystal oscillator is integrated in the microcontroller having crystal frequency of 12 MHz.



- **Pins 1 to 8** − These pins are known as Port 1. This port doesn't serve any other functions. It is internally pulled up, bi-directional I/O port.
- **Pin 9** − It is a RESET pin, which is used to reset the microcontroller to its initial values.
- **Pins 10 to 17** − These pins are known as Port 3. This port serves some functions like interrupts, timer input, control signals, serial communication signals RxD and TxD, etc.

Pin	Name	Bit Address	Function
P3.0	RXD	B0H	Receive data for serial port
P3.1	TXD	B1H	Transmit data for serial port
P3.2	INT0-bar	B2H	External interrupt 0
P3.3	INT1-bar	B3H	External interrupt 1
P3.4	T0	B4H	Timer/counter 0 external input
P3.5	T1	B5H	Timer/counter 1 external input
P3.6	WR-bar	B6H	External data memory write strobe
P3.7	RD-bar	B7H	External data memory read strobe

- **Pins 18 & 19** − These pins are used for interfacing an external crystal to get the system clock.
- **Pin 20** − This pin provides the power supply to the circuit. When the only positive power supply is used then VSS (Voltage Source Supply) means ground or zero.
- **Pins 21 to 28** − These pins are known as Port 2. It serves as I/O port. Higher order address bus signals are also multiplexed using this port.
- **Pin 29** − This is PSEN pin which stands for Program Store Enable. It is used to read a signal from the external program memory.
- **Pin 30** − This is EA pin which stands for External Access input. It is used to enable/disable the external memory interfacing.
- **Pin 31** − This is ALE pin which stands for Address Latch Enable. It is used to demultiplex the address-data signal of port.
- **Pins 32 to 39** − These pins are known as Port 0. It serves as I/O port. Lower order address and data bus signals are multiplexed using this port.
- **Pin 40** − This pin is used to provide power supply to the circuit. VCC (Voltage Common Collector) is the higher voltage with respect to GND (ground). VCC is the power input of a device. It may be positive or negative with respect to GND.

Instruction Set:

As the 8051 family of Microcontrollers are 8-bit processors, the 8051 Microcontroller Instruction Set is optimized for 8-bit control applications. As a typical 8-bit processor, the 8051 Microcontroller instructions have 8-bit Opcodes. As a result, the 8051 Microcontroller instruction set can have up to $2^8 = 256$ Instructions.

A Brief Look at 8051 Microcontroller Instructions and Groups

The following table shows the 8051 Instruction Groups and Instructions in each group. There are 49 Instruction Mnemonics in the 8051 Microcontroller Instruction Set and these 49 Mnemonics are divided into five groups.

DATA TRANSFER	***ARITHMETIC***	***LOGICAL***	***BOOLEAN***	***PROGRAM BRANCHING***
MOV	ADD	ANL	CLR	LJMP
MOVC	ADDC	ORL	SETB	AJMP
MOVX	SUBB	XRL	MOV	SJMP
PUSH	INC	CLR	JC	JZ
POP	DEC	CPL	JNC	JNZ
XCH	MUL	RL	JB	CJNE
XCHD	DIV	RLC	JNB	DJNZ
	DA A	RR	JBC	NOP
		RRC	ANL	LCALL
		SWAP	ORL	ACALL
			CPL	RET
				RETI
				JMP

8051 Addressing Modes

What is an Addressing Mode?

An Addressing Mode is a way to locate a target Data, which is also called as Operand. The 8051 Family of Microcontrollers allows five types of Addressing Modes for addressing the Operands. They are:

- Immediate Addressing
- Register Addressing
- Direct Addressing
- Register – Indirect Addressing
- Indexed Addressing

Immediate Addressing

In Immediate Addressing mode, the operand, which follows the Opcode, is a constant data of either 8 or 16 bits. The name Immediate Addressing came from the fact that the constant data to be stored in the memory immediately follows the Opcode.

The constant value to be stored is specified in the instruction itself rather than taking from a register. The destination register to which the constant data must be copied should be the same size as the operand mentioned in the instruction.

Example: MOV A, #030H

Here, the Accumulator is loaded with 30 (hexadecimal). The # in the operand indicates that it is a data and not the address of a Register.

Immediate Addressing is very fast as the data to be loaded is given in the instruction itself.

Register Addressing

In the 8051 Microcontroller Memory Organization Tutorial, we have seen the organization of RAM and four banks of Working Registers with eight Registers in each bank.

In Register Addressing mode, one of the eight registers (R0 – R7) is specified as Operand in the Instruction. It is important to select the appropriate Bank with the help of PSW Register. Let us see a example of Register Addressing assuming that Bank0 is selected.

Example: MOV A, R5

Here, the 8-bit content of the Register R5 of Bank0 is moved to the Accumulator.

Direct Addressing

In Direct Addressing Mode, the address of the data is specified as the Operand in the instruction. Using Direct Addressing Mode, we can access any register or on-chip variable. This includes general purpose RAM, SFRs, I/O Ports, Control registers.

Example: MOV A, 47H

Here, the data in the RAM location 47H is moved to the Accumulator.

Register Indirect Addressing

In the Indirect Addressing Mode or Register Indirect Addressing Mode, the address of the Operand is specified as the content of a Register. This will be clearer with an example.

Example: MOV A, @R1

The @ symbol indicates that the addressing mode is indirect. If the contents of R1 is 56H, for example, then the operand is in the internal RAM location 56H. If the contents of the RAM location 56H is 24H, then 24H is moved into accumulator.

Only R0 and R1 are allowed in Indirect Addressing Mode. These register in the indirect addressing mode are called as Pointer registers.

Indexed Addressing Mode

With Indexed Addressing Mode, the effective address of the Operand is the sum of a base register and an offset register. The Base Register can be either Data Pointer (DPTR) or Program Counter (PC) while the Offset register is the Accumulator (A).

In Indexed Addressing Mode, only MOVC and JMP instructions can be used. Indexed Addressing Mode is useful when retrieving data from look-up tables.

Example: MOVC A, @A+DPTR

Here, the address for the operand is the sum of contents of DPTR and Accumulator.

Types of Instructions in 8051 Microcontroller Instruction Set

Before seeing the types of instructions, let us see the structure of the 8051 Microcontroller Instruction. An 8051 Instruction consists of an Opcode (short of Operation – Code) followed by Operand(s) of size Zero Byte, One Byte or Two Bytes.

The Op-Code part of the instruction contains the Mnemonic, which specifies the type of operation to be performed. All Mnemonics or the Opcode part of the instruction are of One Byte size.

Coming to the Operand part of the instruction, it defines the data being processed by the instructions. The operand can be any of the following:

- No Operand
- Data value
- I/O Port
- Memory Location
- CPU register

There can multiple operands and the format of instruction is as follows:

MNEMONIC DESTINATION_OPERAND, SOURCE_OPERAND

A simple instruction consists of just the opcode. Other instructions may include one or more operands. Instruction can be one-byte instruction, which contains only opcode, or two-byte instructions, where the second byte is the operand or three byte instructions, where the operand makes up the second and third byte.

Based on the operation they perform, all the instructions in the 8051 Microcontroller Instruction Set are divided into five groups. They are:

- Data Transfer Instructions
- Arithmetic Instructions
- Logical Instructions
- Boolean or Bit Manipulation Instructions
- Program Branching Instructions

We will now see about these instructions briefly.

Data Transfer Instructions

The Data Transfer Instructions are associated with transfer with data between registers or external program memory or external data memory. The Mnemonics associated with Data Transfer are given below.

- *MOV*

- *MOVC*
- *MOVX*

- *PUSH*
- *POP*
- *XCH*
- *XCHD*

The following table lists out all the possible data transfer instruction along with other details like addressing mode, size occupied and number machine cycles it takes.

Mnemonic	Instruction	Description	Addressing Mode	# of Bytes	# of Cycles
MOV	A, #Data	A ← Data	Immediate	2	1
	A, Rn	A ← Rn	Register	1	1
	A, Direct	A ← (Direct)	Direct	2	1
	A, @Ri	A ← @Ri	Indirect	1	1
	Rn, #Data	Rn ← data	Immediate	2	1
	Rn, A	Rn ← A	Register	1	1
	Rn, Direct	Rn ← (Direct)	Direct	2	2
	Direct, A	(Direct) ← A	Direct	2	1
	Direct, Rn	(Direct) ← Rn	Direct	2	2
	Direct1, Direct2	(Direct1) ← (Direct2)	Direct	3	2
	Direct, @Ri	(Direct) ← @Ri	Indirect	2	2
	Direct, #Data	(Direct) ← #Data	Direct	3	2
	@Ri, A	@Ri ← A	Indirect	1	1
	@Ri, Direct	@Ri ← Direct	Indirect	2	2
	@Ri, #Data	@Ri ← #Data	Indirect	2	1
	DPTR, #Data16	DPTR ← #Data16	Immediate	3	2
MOVC	A, @A+DPTR	A ← Code Pointed by A+DPTR	Indexed	1	2
	A, @A+PC	A ← Code Pointed by A+PC	Indexed	1	2
	A, @Ri	A ← Code Pointed by Ri (8-bit Address)	Indirect	1	2
MOVX	A, @DPTR	A ← External Data Pointed by DPTR	Indirect	1	2
	@Ri, A	@Ri ← A (External Data 8-bit Addr)	Indirect	1	2
	@DPTR, A	@DPTR ← A (External Data 16-bit Addr)	Indirect	1	2
PUSH	Direct	Stack Pointer SP ← (Direct)	Direct	2	2
POP	Direct	(Direct) ← Stack Pointer SP	Direct	2	2
XCH	Rn	Exchange ACC with Rn	Register	1	1
	Direct	Exchange ACC with Direct Byte	Direct	2	1
	@Ri	Exchange ACC with Indirect RAM	Indirect	1	1
XCHD	A, @Ri	Exchange ACC with Lower Order Indirect RAM	Indirect	1	1

Arithmetic Instructions

Using Arithmetic Instructions, you can perform addition, subtraction, multiplication and division. The arithmetic instructions also include increment by one, decrement by one and a special instruction called Decimal Adjust Accumulator.

The Mnemonics associated with the Arithmetic Instructions of the 8051 Microcontroller Instruction Set are:

- *ADD*
- *ADDC*
- *SUBB*
- *INC*
- *DEC*
- *MUL*
- *DIV*
- *DA A*

The arithmetic instructions has no knowledge about the data format i.e. signed, unsigned, ASCII, BCD, etc. Also, the operations performed by the arithmetic instructions affect flags like carry, overflow, zero, etc. in the PSW Register.

All the possible Mnemonics associated with Arithmetic Instructions are mentioned in the following table.

Mnemonic	Instruction	Description	Addressing Mode	# of Bytes	# of Cycles
ADD	A, #Data	A ← A + Data	Immediate	2	1
	A, Rn	A ← A + Rn	Register	1	1
	A, Direct	A ← A + (Direct)	Direct	2	1
	A, @Ri	A ← A + @Ri	Indirect	1	1
ADDC	A, #Data	A ← A + Data + C	Immediate	2	1
	A, Rn	A ← A + Rn + C	Register	1	1
	A, Direct	A ← A + (Direct) + C	Direct	2	1
	A, @Ri	A ← A + @Ri + C	Indirect	1	1
SUBB	A, #Data	A ← A – Data – C	Immediate	2	1
	A, Rn	A ← A – Rn – C	Register	1	1
	A, Direct	A ← A – (Direct) – C	Direct	2	1
	A, @Ri	A ← A – @Ri – C	Indirect	1	1
MUL	AB	Multiply A with B (A ← Lower Byte of A*B and B ← Higher Byte of A*B)	--	1	4
DIV	AB	Divide A by B (A ← Quotient and B ← Remainder)	--	1	4
DEC	A	A ← A – 1	Register	1	1
	Rn	Rn ← Rn – 1	Register	1	1
	Direct	(Direct) ← (Direct) – 1	Direct	2	1
	@Ri	@Ri ← @Ri – 1	Indirect	1	1
INC	A	A ← A + 1	Register	1	1
	Rn	Rn ← Rn + 1	Register	1	1
	Direct	(Direct) ← (Direct) + 1	Direct	2	1
	@Ri	@Ri ← @Ri + 1	Indirect	1	1
	DPTR	DPTR ← DPTR + 1	Register	1	2
DA	A	Decimal Adjust Accumulator	--	1	1

Logical Instructions

The next group of instructions are the Logical Instructions, which perform logical operations like AND, OR, XOR, NOT, Rotate, Clear and Swap. Logical Instruction are performed on Bytes of data on a bit-by-bit basis.

Mnemonics associated with Logical Instructions are as follows:

- *ANL*
- *ORL*
- *XRL*
- *CLR*
- *CPL*
- *RL*
- *RLC*
- *RR*
- *RRC*
- *SWAP*

The following table shows all the possible Mnemonics of the Logical Instructions.

Mnemonic	Instruction	Description	Addressing Mode	# of Bytes	# of Cycles
ANL	A, #Data	A ← A AND Data	Immediate	2	1
	A, Rn	A ← A AND Rn	Register	1	1
	A, Direct	A ← A AND (Direct)	Direct	2	1
	A, @Ri	A ← A AND @Ri	Indirect	1	1
	Direct, A	(Direct) ← (Direct) AND A	Direct	2	1
	Direct, #Data	(Direct) ← (Direct) AND #Data	Direct	3	2
ORL	A, #Data	A ← A OR Data	Immediate	2	1
	A, Rn	A ← A OR Rn	Register	1	1
	A, Direct	A ← A OR (Direct)	Direct	2	1
	A, @Ri	A ← A OR @Ri	Indirect	1	1
	Direct, A	(Direct) ← (Direct) OR A	Direct	2	1
	Direct, #Data	(Direct) ← (Direct) OR #Data	Direct	3	2
XRL	A, #Data	A ← A XRL Data	Immediate	2	1
	A, Rn	A ← A XRL Rn	Register	1	1
	A, Direct	A ← A XRL (Direct)	Direct	2	1
	A, @Ri	A ← A XRL @Ri	Indirect	1	1
	Direct, A	(Direct) ← (Direct) XRL A	Direct	2	1
	Direct, #Data	(Direct) ← (Direct) XRL #Data	Direct	3	2
CLR	A	A← 00H	--	1	1
CPL	A	A ← A	--	1	1
RL	A	Rotate ACC Left	--	1	1
RLC	A	Rotate ACC Left through Carry	--	1	1
RR	A	Rotate ACC Right	--	1	1
RRC	A	Rotate ACC Right through Carry	--	1	1
SWAP	A	Swap Nibbles within ACC	--	1	1

Boolean or Bit Manipulation Instructions

As the name suggests, Boolean or Bit Manipulation Instructions will deal with bit variables. We know that there is a special bit-addressable area in the RAM and some of the Special Function Registers (SFRs) are also bit addressable.

The Mnemonics corresponding to the Boolean or Bit Manipulation instructions are:

- *CLR*
- *SETB*
- *MOV*
- *JC*
- *JNC*
- *JB*
- *JNB*
- *JBC*
- *ANL*
- *ORL*
- *CPL*

These instructions can perform set, clear, and, or, complement etc. at bit level. All the possible mnemonics of the Boolean Instructions are specified in the following table.

Mnemonic	Instruction	Description	# of Bytes	# of Cycles
CLR	C	C ← 0 (C = Carry Bit)	1	1
	Bit	Bit ← 0 (Bit = Direct Bit)	2	1
SET	C	C ← 1	1	1
	Bit	Bit ← 1	2	1
CPL	C	C ← $\overline{C}$	1	1
	Bit	Bit ← $\overline{Bit}$	2	1
ANL	C, /Bit	C ← C. $\overline{Bit}$ (AND)	2	1
	C, Bit	C ← C . Bit (AND)	2	1
ORL	C, /Bit	C ← C + $\overline{Bit}$ (OR)	2	1
	C, Bit	C ← C + Bit (OR)	2	1
MOV	C, Bit	C ← Bit	2	1
	Bit, C	Bit ← C	2	2
JC	rel	Jump is Carry (C) is Set	2	2
JNC	rel	Jump is Carry (C) is Not Set	2	2
JB	Bit, rel	Jump is Direct Bit is Set	3	2
JNB	Bit, rel	Jump is Direct Bit is Not Set	3	2
JBC	Bit, rel	Jump is Direct Bit is Set and Clear Bit	3	2

Program Branching Instructions

The last group of instructions in the 8051 Microcontroller Instruction Set are the Program Branching Instructions. These instructions control the flow of program logic. The mnemonics of the Program Branching Instructions are as follows.

- *LJMP*
- *AJMP*
- *SJMP*
- *JZ*
- *JNZ*
- *CJNE*
- *DJNZ*
- *NOP*
- *LCALL*
- *ACALL*
- *RET*
- *RETI*
- *JMP*

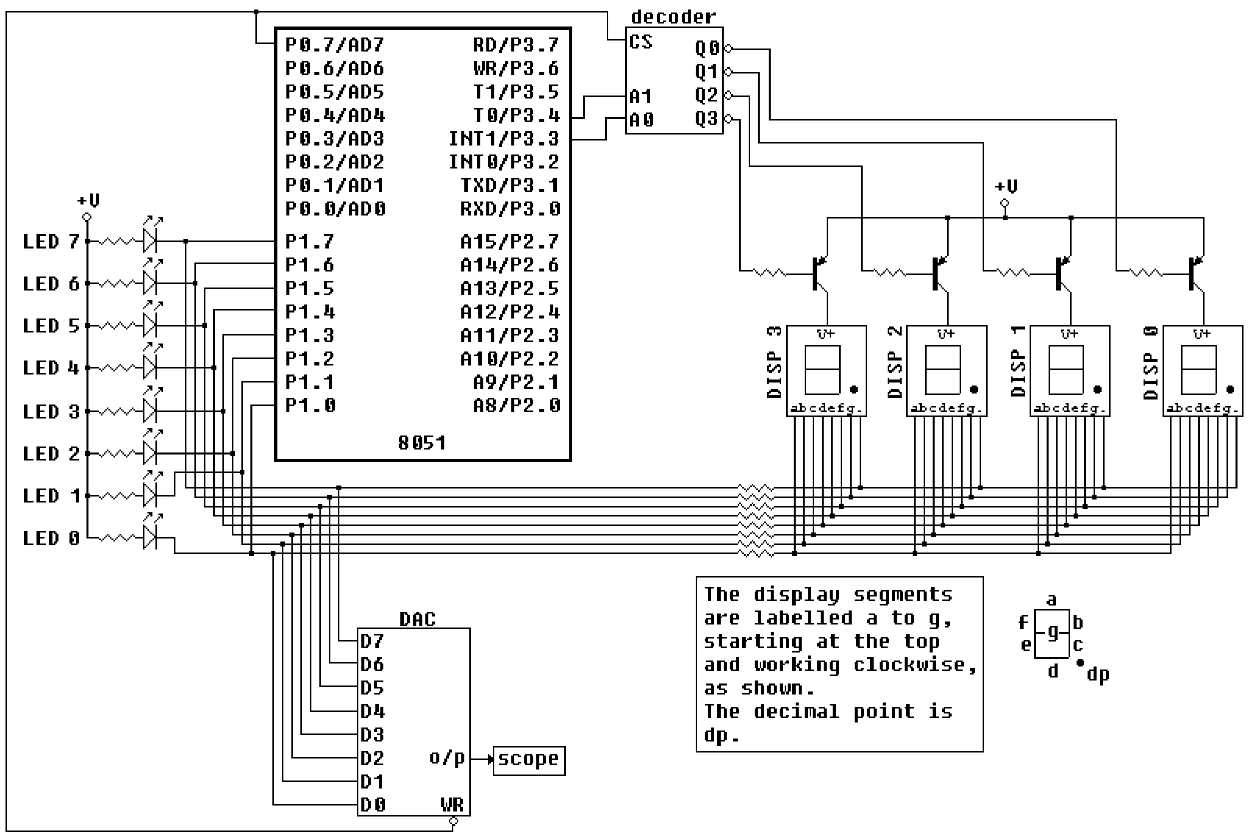
All these instructions, except the NOP (No Operation) affect the Program Counter (PC) in one way or other. Some of these instructions has decision making capability before transferring control to other part of the program.

The following table shows all the mnemonics with respect to the program branching instructions.

Mnemonic	Instruction	Description	# of Bytes	# of Cycles
ACALL	ADDR11	Absolute Subroutine Call PC + 2 → (SP); ADDR11 → PC	2	2
LCALL	ADDR16	Long Subroutine Call PC + 3 → (SP); ADDR16 → PC	3	2
RET	--	Return from Subroutine (SP) → PC	1	2
RETI	--	Return from Interrupt	1	2
AJMP	ADDR11	Absolute Jump ADDR11 → PC	2	2
LJMP	ADDR16	Long Jump ADDR16 → PC	3	2
SJMP	rel	Short Jump PC + 2 + rel → PC	2	2
JMP	@A + DPTR	A + DPTR → PC	1	2
JZ	rel	If A=0, Jump to PC + rel	2	2
JNZ	rel	If A ≠ 0, Jump to PC + rel		
CJNE	A, Direct, rel	Compare (Direct) with A. Jump to PC + rel if not equal	3	2
	A, #Data, rel	Compare #Data with A. Jump to PC + rel if not equal	3	2
	Rn, #Data, rel	Compare #Data with Rn. Jump to PC + rel if not equal	3	2
	@Ri, #Data, rel	Compare #Data with @Ri. Jump to PC + rel if not equal	3	2
			ELECTRONICS HUB	
DJNZ	Rn, rel	Decrement Rn. Jump to PC + rel if not zero	2	2
	Direct, rel	Decrement (Direct). Jump to PC + rel if not zero	3	2
NOP		No Operation	1	1

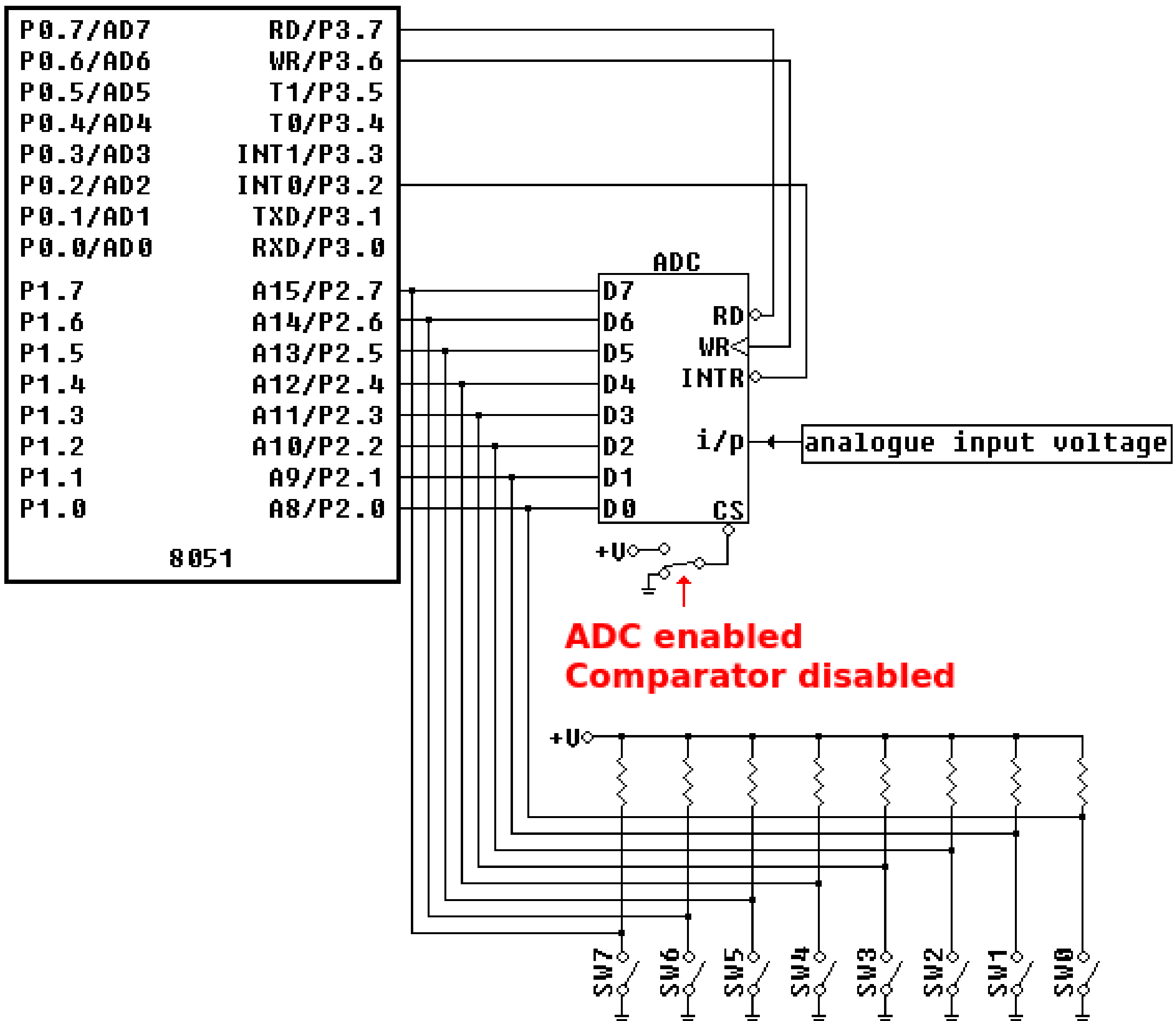
The LED Bank:

As can be seen in the diagram below, the LED bank, the DAC inputs and the 7-segment display data lines all share port 1.



The Switches:

When a switch is open a logic 1 appears on the port pin (via the pull-up resistor) while closing the switch connects the pin directly to ground - logic 0. The switch bank and the outputs of the ADC are applied to port 2.



A Few Notes on the Assembler:

The 2-pass assembler with the EdSim51 Simulator is not a full-blown assembler. It does not link multiple files and only some of the directives you might expect are implemented. However, I feel it is adequate for the beginner. Below is a list of its features:

- All of the 8051 instructions are implemented, except for MOVX instructions, as the simulator does not handle external memory.
- JMP rel equates to either SJMP rel or AJMP rel. LJMP rel must be programmed explicitly.
- Similarly, CALL equates to ACALL. LCALL must be programmed explicitly.
- SET and EQU directives are implemented.
- ORG is implemented.
- USING directive (states which register bank is being used) is implemented.
- ARn equates to the register address, as specified by USING (if the register bank is not specified prior to ARn's use, register bank 0 is assumed).
- SFR names and SFR bit names equate to the appropriate address.
- HIGH followed by an operand in brackets equates to the high byte of the operand.
- LOW followed by an operand in brackets equates to the low byte of the operand.
- Labels are followed by a colon.
- The default for numerical values is decimal. Hex values can be entered by appending H after the number, or placing 0x before it. If H is used, the number cannot begin with a letter (example: F5H must be written as 0F5H). Binary values are entered by appending B after the number (as shown in the image below).
- The assembler is not case-sensitive.

Lab Tasks:

1. Write a program which act as up and down counter from 0 to 9 and 9 to 0 using the 8-LEDs.
2. Write a program which turn on the rightmost led by pressing the leftmost switch and so on.
3. Write a program code which shows the number of key pressed on LED bar.

- **You have to submit task no 2 as your lab report task.**

Lab Performance Evaluation:

Read and practice the manual. Performance is evaluated at the end of lab based on successfully running and understanding of examples, tasks and viva using defined rubrics.

Lab Report Evaluation:

Submit the Lab Report by writing all the examples and tasks which must include the following:

1. Brief description about how its work (3-5 lines)
2. The code
3. The results (Screenshot)

Important note: Lab Report must be according to the Lab Report Format available on Google Classroom and must be submitted on time. Two similar reports will get zero marks. So, don't try to copy your fellow reports. Lab report must be submitted in group of three maximum.